theory and experimental measurements show to be about 186,000 miles per second, or about 670 million miles per hour. This is fast enough to circle the earth more than seven times in a second. If Slim, for example, were to travel not at 120 miles per hour but at 580 million miles per hour (about 87 percent of light speed), the mathematics of special relativity predicts that Jim would measure the length of the car to be about eight feet, which is substantially different from Slim's measurement (as well as the specifications in the owner's manual). Similarly, the time to traverse the drag strip according to Jim will be about twice as long as the time measured by Slim.

Since such enormous speeds are far beyond anything currently attainable, the effects of "time dilation" and "Lorentz contraction," as these phenomena are technically called, are extremely small in day-to-day life. If we happened to live in a world in which things typically traveled at speeds close to that of light, these properties of space and time would be so completely intuitive—since we would experience them constantly—that they would deserve no more discussion than the apparent motion of trees on the side of the road mentioned at the outset of this chapter. But since we don't live in such a world, these features are unfamiliar. As we shall see, understanding and accepting them requires that we subject our worldview to a thorough makeover.

The Principle of Relativity

There are two simple yet deeply rooted structures that form the foundation of special relativity. As mentioned, one concerns properties of light; we shall discuss this more fully in the next section. The other is more abstract. It is concerned not with any specific physical law but rather with all physical laws, and is known as the principle of relativity The principle of relativity rests on a simple fact: Whenever we discuss speed or velocity (an object's speed and its direction of motion), we must specify precisely who or what is doing the measuring. Understanding the meaning and importance of this statement is easily accomplished by contemplating the following situation.

Imagine that George, who is wearing a spacesuit with a small, red flashing light, is floating in the absolute darkness of completely empty space, far away from any planets, stars, or galaxies. From George's perspective, he is completely stationary, engulfed in the uniform, still blackness of the cosmos. Off in the distance, George catches sight of a